

A REPORT WRITING
ON BRINELL HARDNESS TEST

IN
PARTIAL COMPLETION OF
MEE 813
APPLIED MANUFACTURING ENGINEERING METHODS

BY
ADELEKE OLORUNNISOLA OYEYEMI: MEE/13/3858

DEPARTMENT OF MECHANICAL ENGINEERING,
SCHOOL OF ENGINEERING,
FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

AUGUST, 2023

The Brinell Hardness Test: A Foundational Indentation Method in Strength of Materials Engineering

I. INTRODUCTION: WHY HARDNESS TESTING SITS AT THE CENTER OF STRENGTH OF MATERIALS ENGINEERING

Every course in strength of materials begins with the same triad of properties: stress, strain, and the material's resistance to permanent deformation [9]. Tensile testing gives the most direct access to that triad, but it is slow, it destroys the specimen, and it requires a machined coupon of a specific geometry. Hardness testing answers a narrower question, how much does the material resist localized indentation, but it answers it in seconds, on a specimen that survives the test essentially intact, and on material that is often too coarse, too small, or too irregularly shaped to be turned into a tensile coupon at all. This is precisely why hardness testing, not tensile testing, is the property most frequently measured on the shop floor and in incoming-material quality control: castings, forgings, weldments, and heat-treated components are checked for hardness far more often than they are pulled to failure.

Of the indentation hardness methods, Brinell is the oldest in continuous industrial use. It was introduced by the Swedish engineer Johan August Brinell in 1900, at the Paris Exposition, as a practical, quantitative alternative to the qualitative, comparative scratch methods that had been used until then, principally the Mohs scale, which ranks minerals from talc (1) to diamond (10) purely by which one can scratch the other [1], reproduced in Table I. It is included specifically to make the contrast with Brinell hardness explicit: Mohs is ordinal and qualitative, Brinell is a physically defined, load-and-area-based quantity with the dimensions of pressure.

TABLE I
THE MOHS SCALE OF MINERAL HARDNESS [1], AN ORDINAL, COMPARATIVE RANKING RATHER THAN A PHYSICALLY DEFINED PRESSURE, UNLIKE THE BRINELL NUMBER DERIVED IN SECTION IV.

Mineral	Mohs hardness
Talc	1
Gypsum	2
Calcite	3
Fluorite	4
Apatite	5
Orthoclase	6
Quartz	7
Topaz	8
Corundum	9
Diamond	10

The remainder of this report develops the Brinell test as an engineering measurement, not merely a procedure to be followed. Section II derives what the Brinell hardness number physically represents in terms of the material's flow stress. Section III describes the apparatus. Section IV gives the standardized procedure and the correct governing equation.

Section V works a full numerical example. Sections VI–X cover interpretation, error sources, advantages and limitations, applications, and comparison against Vickers and Rockwell testing. Section XI concludes.

II. THE MECHANICS OF INDENTATION HARDNESS

A. Hardness as an Indirect Measure of Flow Stress

Hardness is not a fundamental material property in the way that Young's modulus or density are. It is a response quantity: the size of the permanent impression left by a standardized indenter under a standardized load, converted into a number with units of pressure (force per unit area) [1]. Because the indentation is small relative to the specimen, the material directly beneath the indenter is not in a simple uniaxial stress state as it would be in a tensile test; it is in a complex, three-dimensional, elastic-plastic stress state that resembles a spherical cavity being expanded into an infinite half-space. Despite this complexity, indentation hardness correlates remarkably well with the material's resistance to plastic flow, and this correlation is what makes the test useful.

B. From Elastic Contact to Fully Plastic Flow

When the spherical indenter first touches the surface, the contact is elastic and is described by Hertzian contact theory [8]: the mean contact pressure p_m rises with increasing load while the contact remains elastic, following

$$p_m = \left(\frac{4E^*}{3\pi} \right) \left(\frac{a}{R} \right), \quad (1)$$

where a is the contact radius, R is the indenter radius, and E^* is the reduced (combined indenter and specimen) elastic modulus. As the load increases further, the mean contact pressure reaches a value at which the material beneath the indenter begins to yield, first in a small plastic zone contained by the surrounding elastic material, and then, as the load continues to rise, the plastic zone breaks through to the free surface and the contact becomes fully plastic. In this fully plastic regime, the mean contact pressure saturates at a value that no longer depends strongly on load, and it is this saturated value that a properly conducted Brinell test measures. Tabor's classical result for the fully plastic indentation of a metal by a rigid sphere is

$$p_m = H \approx C \sigma_y, \quad C \approx 3 \quad (2)$$

where σ_y is the material's uniaxial representative flow stress (approximately its yield strength for a non-strain-hardening material) and C , the constraint factor, is close to 3 for most metals under a spherical indenter, reflecting the geometric constraint imposed on plastic flow by the surrounding elastic material [2]. Equation (2) is the single most important physical fact underlying this entire report: *the Brinell hardness number is, to a good approximation, three times the material's flow stress*. It is the reason hardness testing can stand in for a tensile test in situations where pulling a specimen to failure is impractical.

C. Meyer's Law and the Representative Strain

The elastic-plastic transition described above also explains why the Brinell hardness number is not perfectly constant across different applied loads on the same material, an effect known as the indentation size effect. Meyer's empirical law relates the applied load P to the indentation diameter d through a power law,

$$P = k d^n, \quad (3)$$

where k is a material constant and n , the Meyer index, is related to the material's strain-hardening exponent m by approximately $n \approx m + 2$. For a fully non-hardening, perfectly plastic material $n \approx 2$ and the hardness number is independent of load, which is the ideal case the Brinell test standard assumes. Real engineering metals strain-harden, so n is typically slightly greater than 2, and the standard therefore specifies both the load and the ball diameter as part of the reported hardness designation (Section VI) precisely because the measured number is not perfectly load-invariant.

A further useful concept is the representative strain, an approximate single strain value that characterizes the average plastic strain state beneath the indenter,

$$\varepsilon_r \approx 0.2 \frac{d}{D}, \quad (4)$$

where D is the ball diameter. For the diameter ratios permitted by the Brinell standard (Section IV), this places the representative strain in the range of roughly 5 to 12 percent, which is why Brinell hardness correlates most reliably with a material's flow stress in that same moderate-strain regime, rather than with its strength at very small or very large strains.

D. Correlation with Tensile Strength

Because Equation (2) links hardness to flow stress, and flow stress is closely related to tensile strength for many engineering metals, an empirical linear correlation between Brinell hardness and ultimate tensile strength (UTS) is widely used in practice. For plain-carbon and low-alloy steels,

$$\text{UTS (MPa)} \approx 3.45 \times \text{HB}, \quad (5)$$

a relationship that is accurate to within roughly 10 percent for the steels it was calibrated on [3], and is the practical basis for one of the most valuable uses of the test: estimating a component's tensile strength from a hardness measurement taken directly on the finished part, without removing material for a tensile coupon.

III. APPARATUS AND INDENTER

Fig. 1 shows a labeled Brinell hardness testing machine. The essential subsystems are the loading system, which applies a controlled, shock-free force to the indenter through a hydraulic or dead-weight-lever mechanism; the indenter and its holder; the specimen support table, adjustable in height; and the reading microscope or digital optical system used to measure the resulting indentation diameter. Fig. 2

shows a complete modern digital Brinell tester of the type used to generate the calculations in Section V.



Fig. 2. A modern digital Brinell hardness tester with an integrated optical measurement system and load cell.

The indenter is a hardened, ground and polished sphere, most commonly 10 mm in diameter, made of hardened steel for testing materials up to about 444 HBW, or tungsten carbide for harder materials up to about 650 HBW, beyond which the ball itself would deform and the test would no longer be valid [4]. Smaller ball diameters, typically 5 mm or 2.5 mm, are used with proportionally reduced loads to test smaller specimens or thinner sections while keeping the load-to-diameter-squared ratio, and therefore the representative strain, constant, which is the condition for geometrically similar indentations and directly comparable hardness numbers across ball sizes.

IV. STANDARDIZED TEST PROCEDURE AND THE GOVERNING EQUATION

A. Procedure

The standardized Brinell procedure, as codified in ASTM E10, ISO 6506-1, and detailed further in the ASM Handbook's treatment of mechanical testing [4], [5], [10], proceeds as follows.

- 1) The specimen surface is prepared: cleaned of oil, dirt, and oxide, and ground or polished flat and smooth enough that the indentation edge can be resolved optically. A minimum specimen thickness of roughly ten times the expected indentation depth is required so that the support beneath the indentation does not influence the reading (the "anvil effect").
- 2) A ball diameter D and load P are selected for the material class and thickness, following the standardized P/D^2 ratios (typically 30 for steels and cast irons, 10 for copper alloys, 5 or 2.5 for softer non-ferrous metals) so that the resulting indentation diameter falls inside the valid measurement window described below.
- 3) The indenter is brought into contact with the surface, and the full test load is applied gradually and without

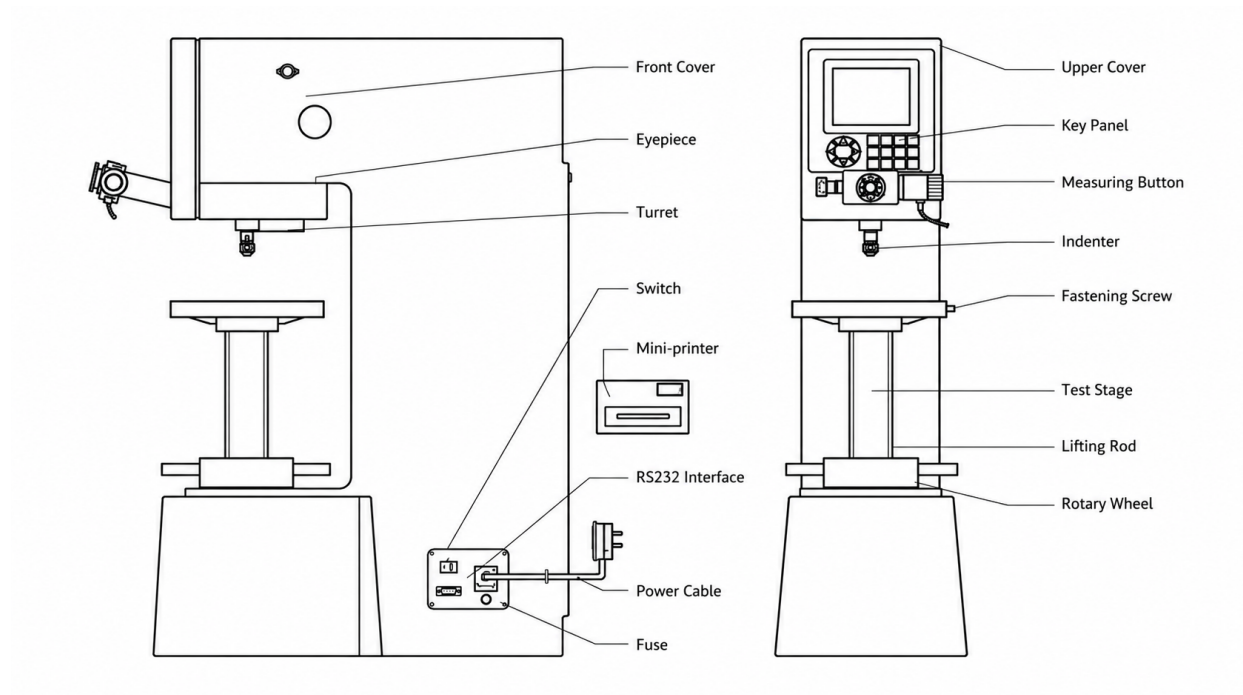


Fig. 1. Labeled front and side view of a Brinell hardness testing machine, showing the load system, indenter and fastening screw, adjustable test stage, and control panel.

shock, avoiding impact loading that would produce an oversized, invalid impression.

- 4) The load is held for a standardized dwell time, typically 10 s to 15 s for steels and up to 30 s for softer, more time-dependent (creeping) materials, allowing the plastic flow beneath the indenter to stabilize before the load is removed.
- 5) The indenter is withdrawn, and the resulting circular indentation is measured optically in two mutually perpendicular directions, d_1 and d_2 , using a calibrated microscope or digital imaging system; the working diameter is $d = (d_1 + d_2)/2$, and any difference greater than about 2% between d_1 and d_2 indicates a non-spherical or off-axis indentation and invalidates the reading.

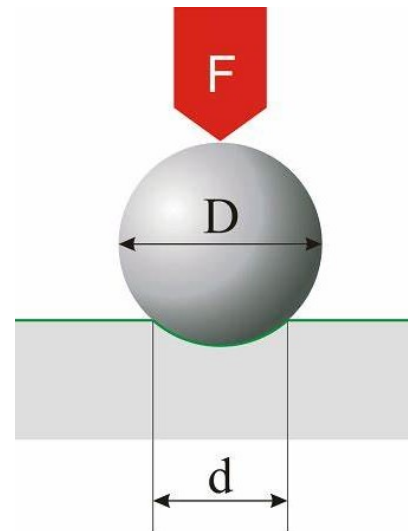


Fig. 3. Idealized schematic of the ball indenter (diameter D) forced under load F into the specimen surface, producing an indentation of diameter d .

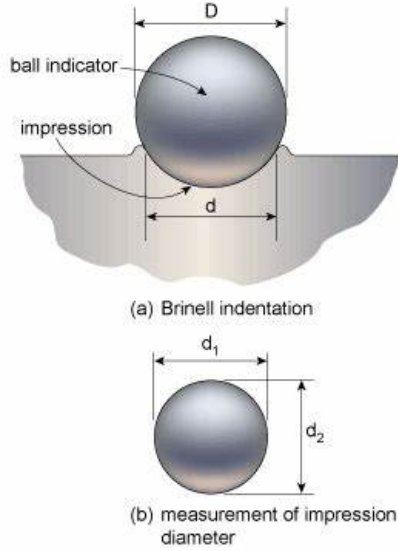


Fig. 4. (a) Cross-section of an actual Brinell indentation, showing the spherical cap geometry the governing equation is derived from. (b) The two-direction diameter measurement, d_1 and d_2 , used to obtain the working diameter d .

B. The Governing Equation, Derived Correctly

The Brinell hardness number is defined as the applied load divided by the *curved surface area of the spherical-cap indentation*, not by the flat projected circular area, a distinction that is frequently, and incorrectly, elided in introductory treatments. Treating the indentation as a spherical cap of chord diameter d cut from a sphere of diameter D , elementary geometry gives the cap's curved surface area as $A = \pi D h$, where h is the cap depth, and h can be eliminated in favor of D and d using $h = \frac{1}{2}(D - \sqrt{D^2 - d^2})$, so that

$$A = \pi D \cdot \frac{1}{2} (D - \sqrt{D^2 - d^2}) = \frac{\pi D}{2} (D - \sqrt{D^2 - d^2}). \quad (6)$$

Dividing the applied load P by this area, and folding the factor of two into the numerator, gives the standard form of the Brinell hardness number,

$$\boxed{\text{HB} = \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})}}, \quad (7)$$

shown graphically in Fig. 5, with P in kilograms-force and D , d in millimeters when the conventional kgf/mm^2 hardness scale is used. Using the curved contact area, rather than the flat projected area $\pi(d/2)^2$, is essential: the two areas differ by roughly 10 to 15 percent over the standard's permitted diameter range, which is far larger than the test's stated repeatability, so a calculation using the projected area would systematically overstate the hardness number by a similar margin.

$$\text{BHN} = \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})}$$

Fig. 5. The Brinell hardness number, correctly defined as applied load divided by the curved spherical-cap surface area of the indentation.

The standard restricts the valid indentation diameter to the range

$$0.24 D \leq d \leq 0.6 D, \quad (8)$$

which corresponds to the representative-strain window discussed in Section II: below $0.24D$ the indentation is too small relative to the ball for the fully plastic Tabor regime to be reliably established and surface-finish effects dominate the reading; above $0.6D$ the flattening of the ball's own curvature against the flat specimen surface begins to violate the spherical-cap assumption the equation is derived from, and the reading again becomes unreliable, particularly for softer materials.

V. WORKED NUMERICAL EXAMPLE

Consider a mild-steel specimen tested with the standard 10 mm ball under the standard steel load of 3000 kgf (29.42 kN), held for 15 s, and suppose the two perpendicular indentation diameters are measured as $d_1 = 4.38$ mm and $d_2 = 4.42$ mm.

First, check validity: d_1 and d_2 agree to within 0.9%, well inside the 2% tolerance, so the impression is acceptably round. The working diameter is

$$d = \frac{4.38 + 4.42}{2} = 4.40 \text{ mm}, \quad (9)$$

which lies inside the required window $0.24D \leq d \leq 0.6D$, i.e., $2.4 \text{ mm} \leq d \leq 6.0 \text{ mm}$, so the reading is valid. Substituting $P = 3000$, $D = 10$, and $d = 4.40$ into Eq. (7):

$$\sqrt{D^2 - d^2} = \sqrt{100 - 19.36} = \sqrt{80.64} = 8.980, \quad (10)$$

$$D - \sqrt{D^2 - d^2} = 10 - 8.980 = 1.020, \quad (11)$$

$$\text{HB} = \frac{2(3000)}{\pi(10)(1.020)} = \frac{6000}{32.04} \approx 187.2. \quad (12)$$

This would be reported as “187 HBW 10/3000/15,” the full designation stating the hardness number, indenter material (tungsten carbide, W), ball diameter, load, and dwell time, exactly as required by ASTM E10 (Section VI). Applying the empirical strength correlation of Eq. (5), this specimen's estimated ultimate tensile strength is $\text{UTS} \approx 3.45 \times 187 \approx 645 \text{ MPa}$, consistent with a medium-carbon steel in a normalized or lightly worked condition, illustrating in one calculation how a fifteen-second indentation test yields both a hardness classification and a usable estimate of tensile strength.

VI. INTERPRETATION OF THE REPORTED HARDNESS NUMBER

Because the measured hardness number depends on the ball diameter, load, and dwell time used to obtain it, the standards (ASTM E10-14, ISO 6506-1:2005) require the full test condition to be reported alongside the number, in the form

[hardness number] HBW [D]/[P]/[H],

where HBW denotes a hardness number obtained with a tungsten-carbide ball (the historical designation HBS or HB referred to now largely obsolete steel-ball tests). For instance, “70 HB 10/500/30” denotes a Brinell hardness of 70 obtained with a 10 mm ball, a 500 kgf load, held for 30 s [2]. This designation is not bureaucratic pedantry: because of the indentation size effect described in Section II, a hardness number reported without its test conditions cannot be safely compared against another number obtained under different conditions.

Table II gives representative Brinell hardness numbers across an enormous span of engineering and non-engineering materials, from soft lead at 5 HB to rhenium diboride, one of the hardest known metal borides, at 4600 HB, illustrating the very wide dynamic range the test can, in principle, address (subject to the indenter-material limits of Section III).

TABLE II
REPRESENTATIVE BRINELL HARDNESS NUMBERS FOR A RANGE OF MATERIALS, SPANNING ALMOST THREE ORDERS OF MAGNITUDE, COMPILED FROM STANDARD PUBLISHED REFERENCE VALUES [3], [10].

Material	Brinell hardness number (HB)
Lead	5.0
Pure aluminium	15
Copper	35
Soft brass	60
Hardened aluminium	75
Mild steel	130
Annealed chisel steel	235
White cast iron	415
Nitrided surface	750
Glass	1550
Rhenium diboride	4600

VII. FACTORS INFLUENCING THE TEST AND SOURCES OF ERROR

Beyond the indentation size effect already discussed, the reliability of a Brinell hardness reading depends on several practical factors that are worth stating explicitly, since they are the actual sources of disagreement between nominally identical tests in practice.

Surface preparation. An unpolished or contaminated surface locally alters the effective contact stiffness and can bias the diameter reading by a fraction of a millimeter, which, given the quadratic dependence of Eq. (7) on d , is enough to shift the computed hardness by several percent.

Specimen and indentation spacing. The standard requires the specimen thickness to be at least eight to ten times the indentation depth, and requires successive indentations to

be separated from each other and from the specimen edge by at least four and two-and-a-half indentation diameters respectively, to avoid the strain field of one indentation, or the free edge, influencing the next reading.

Dwell time and material creep. Materials with significant time-dependent (visco-plastic) behavior continue to deform slowly under the maintained load; too short a dwell time under-registers the indentation for such materials, which is why softer, more creep-prone materials are standardized to a longer, 30-second dwell.

Indenter condition. Wear, flattening, or damage to the ball indenter directly changes the effective contact geometry the governing equation assumes, and the standards require periodic indenter verification against certified reference blocks of known hardness.

Elastic recovery. A small fraction of the indentation depth recovers elastically once the load is removed, meaning the measured indentation is always marginally smaller than the fully loaded impression the underlying mechanics (Section II) describes; this is implicitly absorbed into the empirical constant C in Eq. (2) rather than corrected for explicitly in routine testing.

VIII. ADVANTAGES AND LIMITATIONS

The Brinell test’s principal advantages follow directly from its large indenter and correspondingly large indentation [11]: it is comparatively insensitive to local surface roughness and to microstructural inhomogeneity, because the large indentation averages the material’s response over a volume much larger than a single grain or a single graphite flake in a cast iron, which is precisely why the test remains the preferred method for castings, forgings, and other coarse-grained or heterogeneous materials that would give erratic, unrepresentative readings under a finer indenter such as Vickers. The test is also comparatively simple, inexpensive, and tolerant of a moderately prepared surface.

Its limitations follow from the same large indentation: it is unsuitable for thin sections or small components, where the required minimum thickness cannot be met; it cannot resolve a hardness gradient over a length scale smaller than the indentation itself, so case-hardened or coated surfaces are generally better characterized by Vickers or microhardness methods; and it cannot be used on the hardest engineering materials at all, since above roughly 650 HBW even a tungsten-carbide ball begins to deform, which is the fundamental physical ceiling on the method.

IX. APPLICATIONS ACROSS INDUSTRY

Table III summarizes representative industrial uses. In the aerospace sector, Brinell testing verifies that turbine-blade and structural materials meet their specified hardness, and by extension their estimated strength via Eq. (5), before and after the thermal cycling of service. In the automotive sector, it is used to map hardness distribution along critical rotating components such as crankshafts, where a locally soft region is a fatigue-initiation risk long before it would be visible in any other inspection. In foundries, it is the standard method

for verifying that a heat treatment has produced the intended hardness throughout a casting, precisely because the casting's coarse, as-solidified grain structure is the textbook case for which the Brinell test's large-area averaging was designed.

TABLE III
REPRESENTATIVE INDUSTRIAL APPLICATIONS OF THE BRINELL
HARDNESS TEST.

Industry	Typical application
Automotive	Engine blocks, gears, and crankshafts, where specific hardness levels are required for safety and performance.
Aerospace	Turbine blades, landing gear, and structural elements subject to extreme service loads and temperatures.
Foundry / metallurgy	Verification of uniform hardness and correct heat treatment in castings and forgings.
Tool and die manufacturing	Selection of tool materials and heat-treatment schedules for drill bits, jigs, and welding fixtures.
Mining and heavy machinery	Assessment of hardness of machine components, drilling tools, and processed ore.
R&D / materials science	Screening the hardness of new alloys and materials during development.

X. COMPARISON WITH OTHER INDENTATION HARDNESS METHODS

Table IV places Brinell alongside the two other dominant indentation methods, Vickers and Rockwell, on the basis of an approximate hardness-value correspondence for steels [3]. The Vickers test uses a diamond pyramid indenter under a much smaller load (typically 500 N rather than Brinell's 29.4 kN), and, like Brinell, computes hardness from load divided by indentation surface area, but its small, sharp indentation makes it far better suited to thin sections, coatings, and localized hardness gradients, at the cost of being more sensitive to surface finish and microstructural inhomogeneity than Brinell. The Rockwell test dispenses with optical measurement entirely, inferring hardness directly from indentation *depth* rather than diameter, which makes it fast and operator-independent, at the cost of a result that is less physically transparent and more sensitive to surface preparation and the elastic properties of the underlying support.

TABLE IV
APPROXIMATE CORRESPONDENCE BETWEEN HARDNESS SCALES FOR
STEELS.

Brinell (HB)	Vickers (HV)	Rockwell B	Rockwell C
120	120	67	—
200	200	92	—
240	250	100	22
300	320	—	32
400	420	—	43
500	530	—	51
650	750	—	62

The choice among the three methods in practice is governed almost entirely by specimen geometry and expected hardness gradient, not by any inherent superiority of one method: Brinell for coarse, heterogeneous, or as-cast bulk material; Vickers for thin sections, coatings, and localized or case-hardened regions; Rockwell for rapid, routine production-line screening where speed and operator independence matter more than the physical transparency of the result.

XI. CONCLUSION

The Brinell hardness test converts a fifteen-second, non-destructive indentation into a physically grounded estimate of a material's flow stress, via Tabor's mean-contact-pressure relation, and, through well-established empirical correlations, into a usable estimate of tensile strength, all without machining a tensile coupon or destroying the part under test. Its large spherical indenter, correctly analyzed through the curved spherical-cap contact area rather than the more commonly (and incorrectly) cited flat projected area, gives it a specific and enduring advantage for coarse-grained, heterogeneous, and as-cast materials that persists even in an era of far more sophisticated, computer-controlled hardness testing equipment. Understanding the test at the level developed in this report, from Hertzian elastic contact through fully plastic indentation to the standardized reporting convention, is what separates competently following a testing procedure from genuinely understanding what the resulting number means, and it is that understanding, rather than the procedure alone, that this report has aimed to convey.

REFERENCES

- [1] "Hardness," *Nuclear Power for Everybody*, Materials Science reference. [Online]. Available: <https://www.nuclear-power.com/nuclear-engineering/materials-science/material-properties/hardness/>
- [2] J. Edwards, "Hardness Testing," in *Characterization of Materials*, John Wiley & Sons, 2012, pp. 1–9, doi: 10.1002/0471266965.com025.pub2.
- [3] S. Kalpakjian and S. R. Schmid, *Manufacturing Engineering and Technology*, 8th ed. in SI Units. Pearson, 2018.
- [4] ASTM E10-18, *Standard Test Method for Brinell Hardness of Metallic Materials*, ASTM International, West Conshohocken, PA, 2018.
- [5] ISO 6506-1:2014, *Metallic materials — Brinell hardness test — Part 1: Test method*, International Organization for Standardization, Geneva, 2014.
- [6] ISO 14577-1:2015, *Metallic materials — Instrumented indentation test for hardness and materials parameters — Part 1: Test method*, International Organization for Standardization, Geneva, 2015.
- [7] D. Tabor, *The Hardness of Metals*. Oxford: Clarendon Press, 1951.
- [8] K. L. Johnson, *Contact Mechanics*. Cambridge: Cambridge University Press, 1985.
- [9] W. D. Callister and D. G. Rethwisch, *Materials Science and Engineering: An Introduction*, 9th ed. Hoboken, NJ: John Wiley & Sons, 2014.
- [10] ASM International, *ASM Handbook, Volume 8: Mechanical Testing and Evaluation*. Materials Park, OH: ASM International, 2000.
- [11] H. Chandler (Ed.), *Hardness Testing*, 2nd ed. Materials Park, OH: ASM International, 1999.